

Stage 1 of `did_multiplegt_dyn`:

The math behind the `controls()` pre-estimation block

Notes on lines L894–L1098 of `did_multiplegt_dyn.ado`

Goal of Stage 1. Estimate, separately for each baseline-treatment level d , a slope vector $\hat{\theta}_d \in \mathbb{R}^K$ describing the linear relationship between the outcome's first difference $\Delta Y_{g,t}$ and the controls' first differences $\Delta X_{k,g,t}$, partialling out time fixed effects (interacted with the `trends_nonparam` stratum). Build the matrix Den_d^{-1} used later in the variance correction. Save the per-cell residuals $\widetilde{\Delta X}_{k,g,t}$ that feed the score in `_sumk,d` in Stages 3–4.

1 Notation

$g \in \{1, \dots, G\}$	group index (<code>group_XX</code>).
$t \in \{1, \dots, T\}$	time index (<code>time_XX</code>).
$Y_{g,t}$	outcome (<code>outcome_XX</code>); $\Delta Y_{g,t} = Y_{g,t} - Y_{g,t-1}$ is <code>diff_y_XX</code> .
$D_{g,t}$	treatment; $D_{g,1}$ is the period-one (baseline) treatment, denoted <code>d_sq_XX</code> (or <code>d_sq_int_XX</code> when discretized to integer levels $1, \dots, L_D$).
$d \in \{1, \dots, L_D\}$	a value of the baseline treatment. θ_d is estimated once per d .
$X_{1,g,t}, \dots, X_{K,g,t}$	user-supplied controls (<code>'controls'</code>). $\Delta X_{k,g,t}$ is <code>diff_xk_XX</code> .
$\Delta \mathbf{X}_{g,t} \in \mathbb{R}^K$	vector of controls' first differences, $(\Delta X_{1,g,t}, \dots, \Delta X_{K,g,t})'$.
F_g	first time g 's treatment differs from $D_{g,1}$ (<code>F_g_XX</code>); $F_g = T + 1$ if g never switches.
$N_{g,t}$	weight of cell (g, t) (<code>N_gt_XX</code>). Equals 1 with no weights and group×time data; otherwise the sum of user weights, set to 0 when $Y_{g,t}$ or $D_{g,t}$ is missing.
s_g	the <code>trends_nonparam</code> stratum of g (time-invariant).
G	total number of groups (<code>G_XX</code>).

2 Step 1.1 — Build first differences [ado L902–915]

For each control X_k and each (g, t) :

$$\Delta X_{k,g,t} = X_{k,g,t} - X_{k,g,t-1}. \quad (1)$$

The outcome FD $\Delta Y_{g,t} = Y_{g,t} - Y_{g,t-1}$ has been generated earlier (L891).

Joint non-missing flag. Define

$$\Phi_{g,t} := \prod_{k=1}^K \mathbb{1}\{\Delta X_{k,g,t} \text{ is non-missing}\}, \quad (2)$$

stored as `fd_X_all_non_missing_XX`. Observations with $\Phi_{g,t} = 0$ are excluded from the conditional means below [ado L914].

3 Step 1.2 — The estimation sample $\mathcal{S}_d$ solo quiero aquellos para el mismo $D_{g,1}$ que aun no tengan su primer switch

A cell (g, t) enters the auxiliary regression for baseline level d if and only if:

$$\mathcal{S}_d = \{(g, t) : D_{g,1} = d, \ t < F_g, \ \Delta Y_{g,t} \neq \text{miss}, \ \Phi_{g,t} = 1\}. \quad (3)$$

The constraint $t < F_g$ is the “not-yet-switcher” restriction (`ever_change_d_XX==0&time_XX<F.g_XX`) – it identifies the control sample used to estimate θ_d .

4 Step 1.3 — Conditional mean of $\Delta X_{k,g,t}$ by (d, t, s) [ado L931–934]

For each baseline d , time t , and stratum s , define the weighted sample mean of $\Delta X_{k,g,t}$ over $\mathcal{S}_d$:

$$\overline{\Delta X}_k(d, t, s) = \frac{\sum_{g:(g,t) \in \mathcal{S}_d, \ s_g=s} N_{g,t} \Delta X_{k,g,t}}{\sum_{g:(g,t) \in \mathcal{S}_d, \ s_g=s} N_{g,t}}. \quad (4)$$

Divided by its own weights

In Stata this is the two-line `bys time_XX d_sq_XX 'trends_nonparam' : gegen avg_diff_Xk_XX = total(N_gt_XX * diff_Xk_XX) / sum.weights_control_XX`.

5 Step 1.4 — Residualized FD of controls [ado L941]

The within- (d, t, s) deviation, scaled by $\sqrt{N_{g,t}}$:

$$\widetilde{\Delta X}_{k,g,t} = \sqrt{N_{g,t}} (\Delta X_{k,g,t} - \overline{\Delta X}_k(D_{g,1}, t, s_g)) \cdot \mathbb{I}\{(g, t) \in \mathcal{S}_{D_{g,1}}\}. \quad (5)$$

x_c ~ time fixed effects
obteniendo el residual

(Missing values are replaced by zero [ado L943].)

This is the Frisch–Waugh–Lovell residual of $\Delta X_{k,g,t}$ on time fixed effects (interacted with the `trends_nonparam` stratum), in the weighted sample $\mathcal{S}_d$. The $\sqrt{N_{g,t}}$ factor is the standard trick to convert weighted least squares with weights $N_{g,t}$ into ordinary least squares on the scaled variables.

Define analogously the scaled outcome FD [ado L955]:

$$\widetilde{\Delta Y}_{g,t} = \sqrt{N_{g,t}} \Delta Y_{g,t}. \quad (6)$$

6 Step 1.5 — Per-control score variable [ado L958]

For each k define

$$\text{prod_X}_{k,g,t} := \sqrt{N_{g,t}} \cdot \widetilde{\Delta X}_{k,g,t} = N_{g,t} (\Delta X_{k,g,t} - \overline{\Delta X}_k(D_{g,1}, t, s_g)). \quad (7)$$

This object is not used inside Stage 1 itself; it is the building block of the per-group score in `_sum_{k,d}` that enters $U^{\text{var}, X}$ in Stages 3–4.

7 Step 1.6 — OLS for $\hat{\theta}_d$ via matrix accum [\[ado L1018–1034\]](#)

Stack the residualized controls into a vector

$$\widetilde{\Delta X}_{g,t} := (\widetilde{\Delta X}_{1,g,t}, \dots, \widetilde{\Delta X}_{K,g,t})^\top \in \mathbb{R}^K.$$

The cross-product matrix. `matrix accum overall_XX = diff_y_wXX mycontrols_XX` builds the $(K+1) \times (K+1)$ matrix

$$\mathbf{A}_d = \sum_{(g,t) \in \mathcal{S}_d} \begin{pmatrix} \widetilde{\Delta Y}_{g,t} \\ \widetilde{\Delta X}_{g,t} \end{pmatrix} \begin{pmatrix} \widetilde{\Delta Y}_{g,t} \\ \widetilde{\Delta X}_{g,t} \end{pmatrix}^\top = \begin{pmatrix} \widetilde{Y}^\top \widetilde{Y} & \widetilde{Y}^\top \widetilde{X} \\ \widetilde{X}^\top \widetilde{Y} & \widetilde{X}^\top \widetilde{X} \end{pmatrix}. \quad (8)$$

The code slices out the two blocks needed for OLS:

$$\mathbf{M}_d := \widetilde{X}^\top \widetilde{X} = \text{didmgt_XX} = \text{overall_XX}[2..K+1, 2..K+1], \quad (9)$$

$$\mathbf{q}_d := \widetilde{X}^\top \widetilde{Y} = \text{didmgt_Xy} = \text{overall_XX}[2..K+1, 1]. \quad (10)$$

The estimator.

$$\boxed{\hat{\theta}_d = \mathbf{M}_d^{-1} \mathbf{q}_d = \left(\sum_{(g,t) \in \mathcal{S}_d} \widetilde{\Delta X}_{g,t} \widetilde{\Delta X}_{g,t}^\top \right)^{-1} \sum_{(g,t) \in \mathcal{S}_d} \widetilde{\Delta X}_{g,t} \widetilde{\Delta Y}_{g,t}.} \quad (11)$$

Stored as `coefs_sq_1_XX`, one $K \times 1$ matrix per baseline level $l = d$.

Equivalence with weighted OLS plus time fixed effects (FWL)

The matrix-accum estimator (??) is numerically identical to

$$\hat{\theta}_d = \arg \min_{\theta, \{\alpha_t^{(d,s)}\}} \sum_{(g,t) \in \mathcal{S}_d} N_{g,t} \left(\Delta Y_{g,t} - \Delta X_{g,t}^\top \theta - \alpha_t^{(d,s_g)} \right)^2, \quad (12)$$

i.e. the weighted OLS of $\Delta Y_{g,t}$ on $\Delta X_{g,t}$ plus a full set of time fixed effects $\alpha_t^{(d,s)}$ that are allowed to vary by baseline cell d and `trends_nonparam` stratum s . Equivalence follows from the Frisch–Waugh–Lovell theorem applied to the partial-out in (??)–(??).

The companion regression `reg diff_y_XX 'controlsXX' ibn.time_XX [aw=N_gt_XX] if d_sq_int_XX==d & time_XX<F_g_XX, noconst` [\[ado L984–991\]](#) is run for a different purpose – to produce the fitted values

$$\hat{E}_{g,t}^{(d)} := \widehat{\mathbb{E}} \left[\Delta Y_{g,t} \mid t, s_g, (g,t) \in \mathcal{S}_d \right] \rightsquigarrow \text{E_y_hat_gt_int_‘d’_XX}, \quad (13)$$

which Stages 3–4 use inside `in_sumk,d` to demean $\Delta Y_{g,t}$ within (d, t, s) cells (the $(\Delta Y_{g,t} - E_{y,gt})$ factor at L4664). With `trends_nonparam` present, the regression uses `ibn.time_XX#ibn.trends_nonparam.temp_XX` so $\hat{E}_{g,t}^{(d)}$ is the within- (d, t, s) mean of $\Delta Y_{g,t}$.

8 Step 1.7 — The variance-denominator matrix Den_d^{-1} [\[ado L1052\]](#)

Define the total weight in the variance window for baseline d :

$$N_d^{\text{tot}} := \sum_{(g,t)} N_{g,t} \cdot \mathbb{1} \left\{ 2 \leq t \leq \max_h F_h - 1, \quad t < F_g, \quad \Delta Y_{g,t} \neq \text{miss}, \quad D_{g,1} = d \right\}. \quad (14)$$

The code computes this in two lines [ado L1049–1050]: `gen N_c_‘1’_temp_XX = ...; sum N_gt_XX if N_c_‘1’_temp_XX == 1; then  $N_d^{\text{tot}} = r(\text{sum})$ .`

The matrix.

$$\text{Den}_d^{-1} := \mathbf{M}_d^{-1} \cdot N_d^{\text{tot}} \cdot G, \quad (15)$$

stored as `inv_Denom_‘1’_XX`. The scalings N_d^{tot} and G make Den_d^{-1} comparable across cells when it is multiplied by the per-group scores in the variance formula.

9 Step 1.8 — Singularity guard [ado L1042–1090]

Compute $\det(\mathbf{M}_d)$. If $|\det(\mathbf{M}_d)| \leq 10^{-16}$, flag baseline d in `store_singular_XX` and emit the user warning at L1086–L1090. Two reasons can trigger this:

1. Too few not-yet-switcher observations in cell d relative to K (under-determined OLS).
2. Perfect collinearity among the controls inside $\mathcal{S}_d$ – e.g. a time-invariant covariate has a first difference of zero everywhere.

When this occurs, baseline cell d keeps $\hat{\boldsymbol{\theta}}_d = \mathbf{0}$ and a separate local `store_noresidualization_XX` causes those observations to be *dropped* from estimation [ado L1094–1096].

10 Step 1.9 — The “useful residualization” gate [ado L1003–1010]

Before performing all of the above, the code checks

$$\text{useful_res}_d := \#\{F_g : (g, \cdot) \in \mathcal{S}_d\} > 1, \quad (16)$$

i.e. there exist at least two distinct switching dates in cell d (computed as the number of F_g levels in `tab F_g_XX if d_sq_int_XX==d`). If $\text{useful_res}_d \leq 1$, no residualization is performed for that cell, observations are dropped, and a console message is printed.

11 The Stage 1 output, in one box

For each baseline level d such that $\text{useful_res}_d > 1$ and $\det(\mathbf{M}_d) \neq 0$, Stage 1 returns:

- the slope vector $\hat{\boldsymbol{\theta}}_d \in \mathbb{R}^K$ – `coefs_sq_‘1’_XX`;
- the variance-denominator matrix $\text{Den}_d^{-1} \in \mathbb{R}^{K \times K}$ – `inv_Denom_‘1’_XX`;
- the per-control score series `prod_Xk,g,t – prod_Xk_Ngt_XX`;
- the cell-mean fitted values $\hat{E}_{g,t}^{(d)}$ – `E_y_hat_gt_int_‘d’_XX`;
- the residualized controls $\widetilde{\Delta X}_{k,g,t}$ – `resid_Xk_time_FE_XX` (already multiplied by $\sqrt{N_{g,t}}$).

Stages 2–4 consume these and never recompute them.

12 Compact algorithm (pseudo-code)

```

INPUT  : balanced panel of (Y, X1..XK, D, weights Ngt) by (g,t),
         baseline treatment D{g,1}, trends_nonparam stratum sg, Fg
OUTPUT : { thetad, Dend{-1}, Ehatgt(d), prodXk, residXk } for each d

For k = 1..K :
  DXkgt  <- X{k,g,t} - X{k,g,t-1}
  Phigt   <- AND over k of (DXkgt is non-missing)

For each baseline level d :
  Sd <- { (g,t) : D{g,1}=d, t<Fg, DYgt != miss, Phigt=1 }
  if #{Fg : (g,*) in Sd} <= 1 :
    drop d ; continue

# conditional mean & residual of each control
For k = 1..K :
  avg_DXk(d,t,s) <- weighted mean of DXkgt over (g : (g,t) in Sd, sg=s)
  resid_Xkgt    <- sqrt(Ngt) * (DXkgt - avg_DXk(d,t,sg))
  prod_Xkgt     <- sqrt(Ngt) * resid_Xkgt

# weighted regression
Ytildegt <- sqrt(Ngt) * DYgt
Md <- sum_{Sd} resid_X * resid_X'
qd <- sum_{Sd} resid_X * Ytilde
if |det(Md)| <= 1e-16 :
  flag d as singular ; drop d ; continue
thetad <- Md{-1} * qd

# variance-denominator matrix
Ntotd <- sum Ngt over (g,t) with D{g,1}=d, 2<=t<=max(F)-1, t<Fg, DYgt!=miss
Dend{-1} <- Md{-1} * Ntotd * G

# store cell-mean fitted value of DY (used in Stages 3-4)
Ehatgt(d) <- weighted mean of DYgt over (g : (g,t) in Sd, sg=sg)

```

13 Cross-reference table

Math symbol	.ado variable	First defined at
$\Delta X_{k,g,t}$	diff_Xk_XX	L913
$\Phi_{g,t}$	fd_X_all_non_missing_XX	L900, L914
$\mathcal{S}_d$	(sample filter, not a variable)	L931, L1014
$\overline{\Delta X}_k(d, t, s)$	avg_diff_Xk_XX	L932–934
$\widetilde{\Delta X}_{k,g,t}$	resid_Xk_time_FE_XX	L941
$\widetilde{\Delta Y}_{g,t}$	diff_y_wXX	L955
$\text{prod}_X X_{k,g,t}$	prod_Xk_Ngt_XX	L958
$\mathbf{A}_d$	overall_XX	L1018
$\mathbf{M}_d = \widetilde{X}^\top \widetilde{X}$	didmgt_XX	L1030
$\mathbf{q}_d = \widetilde{X}^\top \widetilde{Y}$	didmgt_Xy	L1031
$\widehat{\boldsymbol{\theta}}_d$	coefs_sq_‘1’_XX	L1034
$\det(\mathbf{M}_d)$	det_XX	L1040
N_d^{tot}	r(sum) from L1050	L1049–1050
Den_d^{-1}	inv_Denom_‘1’_XX	L1052
$\widehat{E}_{g,t}^{(d)}$	E_y_hat_gt_int_‘1’_XX	L996
useful_res_d	useful_res_‘1’_XX	L1009
singular flag	store_singular_‘1’_XX	L1006, L1043